

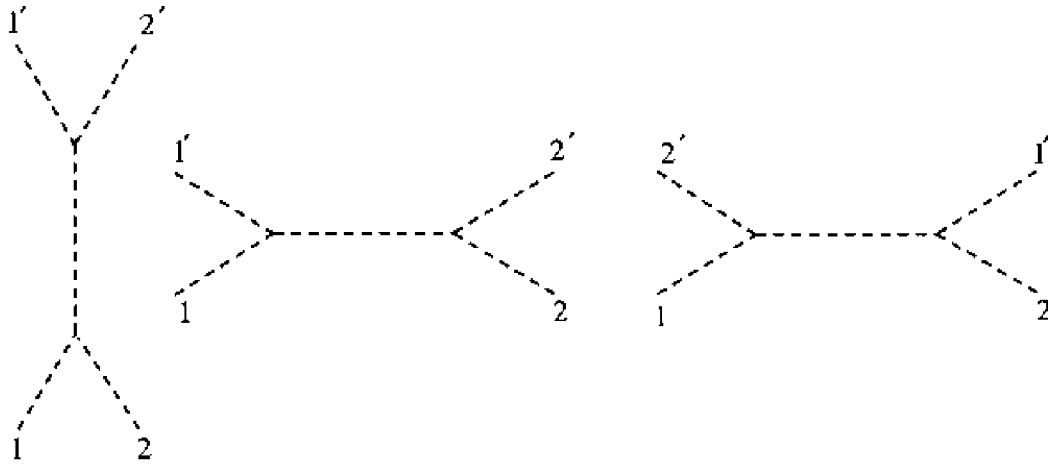


Figure 6.8. The connected-second order diagrams for boson-boson scattering in a theory with interaction (6.1.29).

diagrams is

$$\begin{aligned}
 & S_{\mathbf{p}'_1 \sigma'_1 n'_1 \mathbf{p}'_2 \sigma'_2 n'_2, \mathbf{p}_1 \sigma_1 n_1 \mathbf{p}_2 \sigma_2 n_2} \\
 &= (-i)^2 (2\pi)^{-6} \sum_{\ell \ell' \ell'' m m' m''} g_{\ell \ell' \ell''} g_{m m' m''} \int d^4 x \int d^4 y \left(-i \Delta_{\ell'' m''}(x, y) \right) \\
 &\quad \times \left[u_{\ell}^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-i p'_1 \cdot x} u_{\ell'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-i p'_2 \cdot x} \right. \\
 &\quad \quad \times u_m(\mathbf{p}_1 \sigma_1 n_1) e^{i p_1 \cdot y} u_{m'}(\mathbf{p}_2 \sigma_2 n_2) e^{i p_2 \cdot y} \\
 &\quad + u_{\ell'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-i p'_1 \cdot x} u_{\ell}(\mathbf{p}_1 \sigma_1 n_1) e^{i p_1 \cdot x} \\
 &\quad \quad \times u_{m'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-i p'_2 \cdot y} u_m(\mathbf{p}_2 \sigma_2 n_2) e^{i p_2 \cdot y} \\
 &\quad + u_{\ell'}^*(\mathbf{p}'_2 \sigma'_2 n'_2) e^{-i p'_2 \cdot x} u_{\ell}(\mathbf{p}_1 \sigma_1 n_1) e^{i p_1 \cdot x} \\
 &\quad \quad \times u_{m'}^*(\mathbf{p}'_1 \sigma'_1 n'_1) e^{-i p'_1 \cdot y} u_m(\mathbf{p}_2 \sigma_2 n_2) e^{i p_2 \cdot y} \left. \right]. \quad (6.1.30)
 \end{aligned}$$

To be even more specific, if the bosons in this theory are spinless particles of a single species, then we write the interaction (6.1.29) in the form

$$\mathcal{H} = g \phi^3 / 3! \quad (6.1.31)$$

and the S -matrix element (6.1.30) for scalar-scalar scattering is

$$\begin{aligned}
 & S_{\mathbf{p}'_1 \mathbf{p}'_2, \mathbf{p}_1 \mathbf{p}_2} = \\
 & \frac{ig^2}{(2\pi)^6 \sqrt{16 E'_1 E'_2 E_1 E_2}} \int d^4 x \int d^4 y \Delta_F(x - y) \\
 & \quad \times \left[\exp(-i(p'_1 + p'_2) \cdot x) \exp(i(p_1 + p_2) \cdot y) \right. \\
 & \quad + \exp(i(p_1 - p'_1) \cdot x) \exp(i(p_2 - p'_2) \cdot y) \\
 & \quad \left. + \exp(i(p_1 - p'_2) \cdot x) \exp(i(p_2 - p'_1) \cdot y) \right],
 \end{aligned}$$